605.486 – Android Development

Module 2e
Layouts

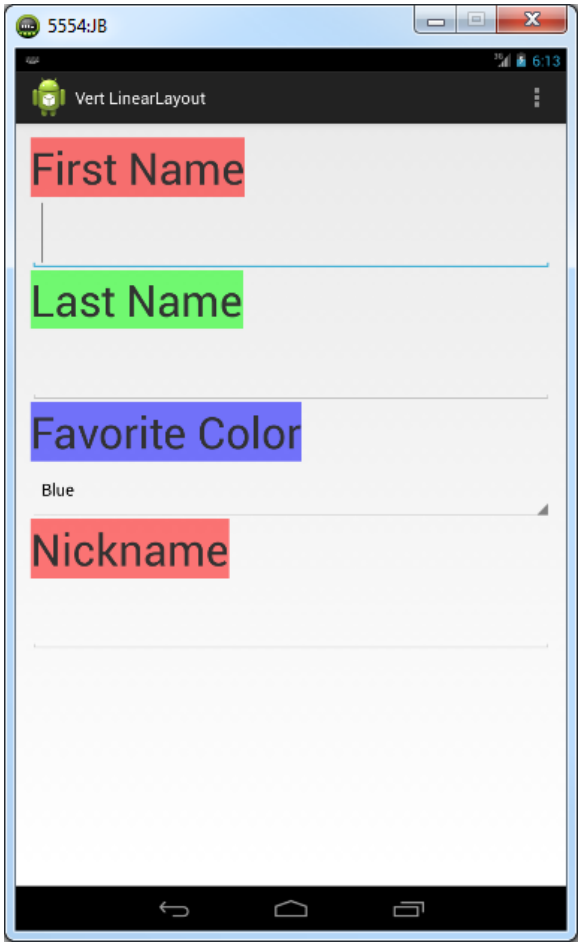
Screen Density

- ldpi : mdpi : hdpi : xhdpi : xxhdpi
- 3 : 4 : 6 : 8 : 12
- 120dpi : 160dpi : 240 dpi : 320dpi : 480dpi
- 1dip = 1 pixel on 160dpi (mdpi) screen
- 1sp = scales based on user-selected font size

Linear Layout

- LinearLayout
  - orientation - horizontal/vertical
  - layout_width / layout_height
    - match_parent
    - wrap_content
    - Explicit amounts – use “dp”
  - layout_weight (more later)
- Alternate layouts (eg: layout-land)

Vertical LinearLayout



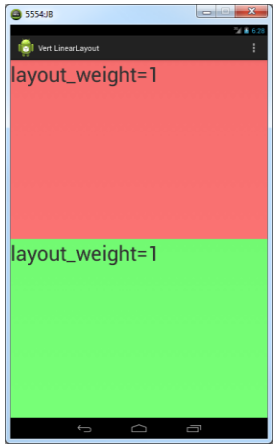

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/first_name" />

    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content">
    </EditText>
    …
```

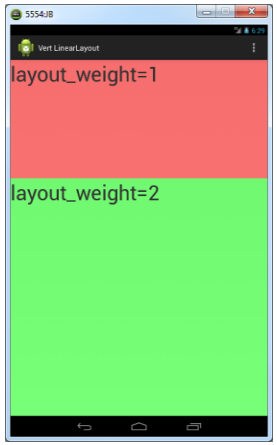
Horizontal LinearLayout (visual)

Horizontal LinearLayout (xml)

```
<LinearLayout android:orientation="vertical"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <LinearLayout android:orientation="horizontal"
    android:layout_width="match_parent"
    android:layout_height="wrap_content">

    <TextView android:text="@string/first_name"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content“ />

    <EditText
      android:layout_width="match_parent"
      android:layout_height="wrap_content"/>
  </LinearLayout>
  …
```

Layout Weight





- For Horizontal layout
  - layout_width=0dp
  - layout_height=wrap_content or #dp
- For Vertical layout
  - layout_height=0dp
  - layout_width=wrap_content or #dp
- layout_weight distributes remaining space in orientation proportionally

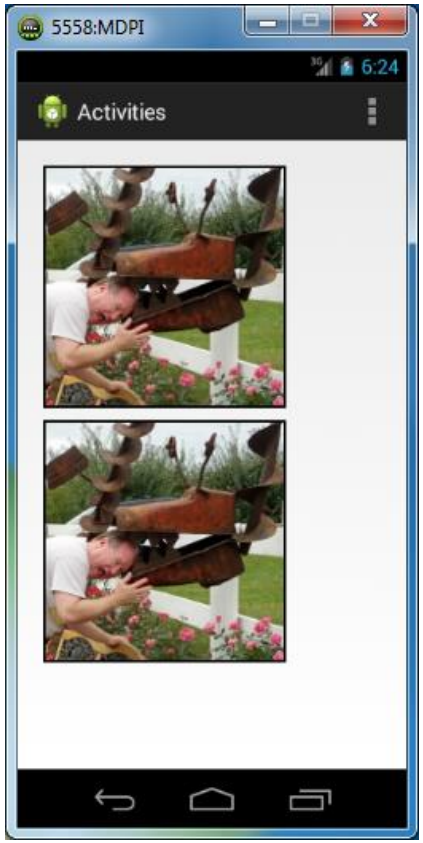
Layout Weight Example

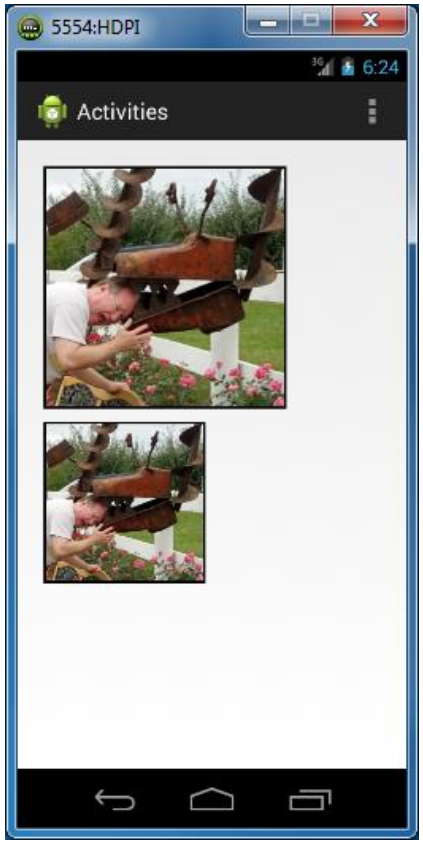
```
<LinearLayout android:orientation="vertical"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView  android:text="@string/layout_weight_1"
      android:layout_width="match_parent"
      android:layout_height="0dp"
      android:layout_weight="1" />

   <TextView android:text="@string/layout_weight_2"
      android:layout_width="match_parent"
      android:layout_height="0dp"
      android:layout_weight="2" />

</LinearLayout>
```

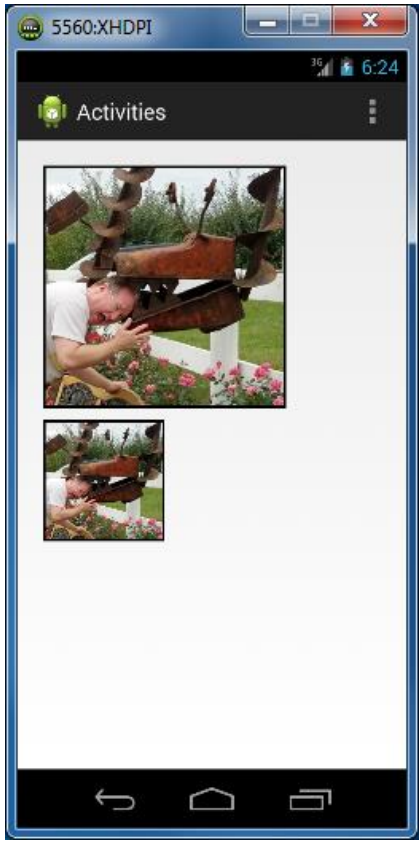
Screen Density

200dpi

200px





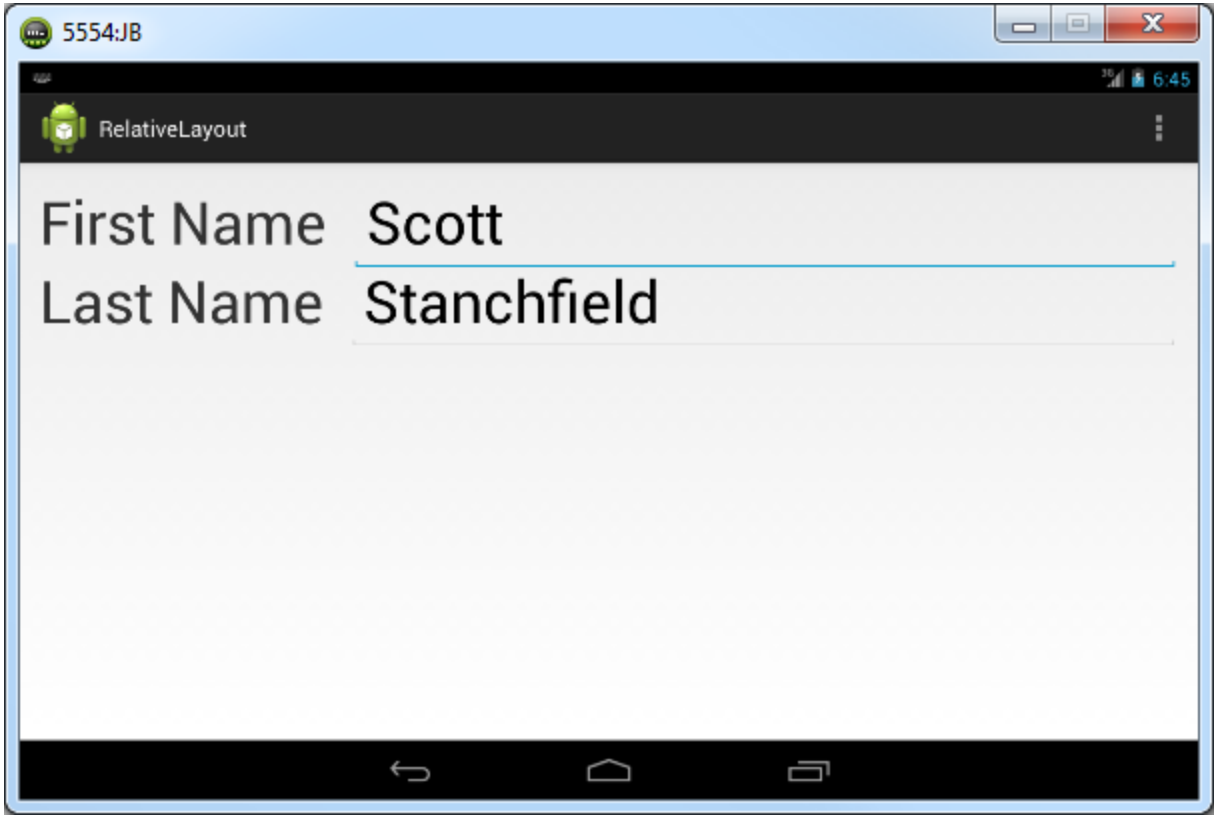


LDPI MDPI HDPI XHDPI

Relative Layout

- Child view relative to parent
  - layout_alignParentLeft (Top, Bottom, Right)
  - layout_centerHorizonal (Vertical, InParent)
- Child view relative to another child
  - layout_alignLeft (Top, Bottom, Right)
  - layout_above (below)
  - layout_toLeftOf (toRightOf)

RelativeLayout Example



RelativeLayout xml (1)

```
<RelativeLayout
  android:layout_width="match_parent“
  android:layout_height="match_parent">
  <TextView android:id="@+id/textView1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentLeft="true"
    android:layout_alignParentTop="true"/>

  <EditText
    android:id="@+id/editText1"
    android:layout_marginLeft="20dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentRight="true"
    android:layout_alignBaseline="@+id/textView1"
    android:layout_toRightOf="@+id/textView1"/>
```

RelativeLayout xml (2)

```
<TextView android:text="@string/last_name"
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@+id/textView1"
    android:layout_alignLeft="@+id/textView1"/>

<EditText
    android:id="@+id/editText1"
    android:layout_marginLeft="20dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentRight="true"
    android:layout_alignBaseline="@+id/textView2"
    android:layout_toRightOf="@+id/textView2"/>

</RelativeLayout>
```

Other Common Layouts

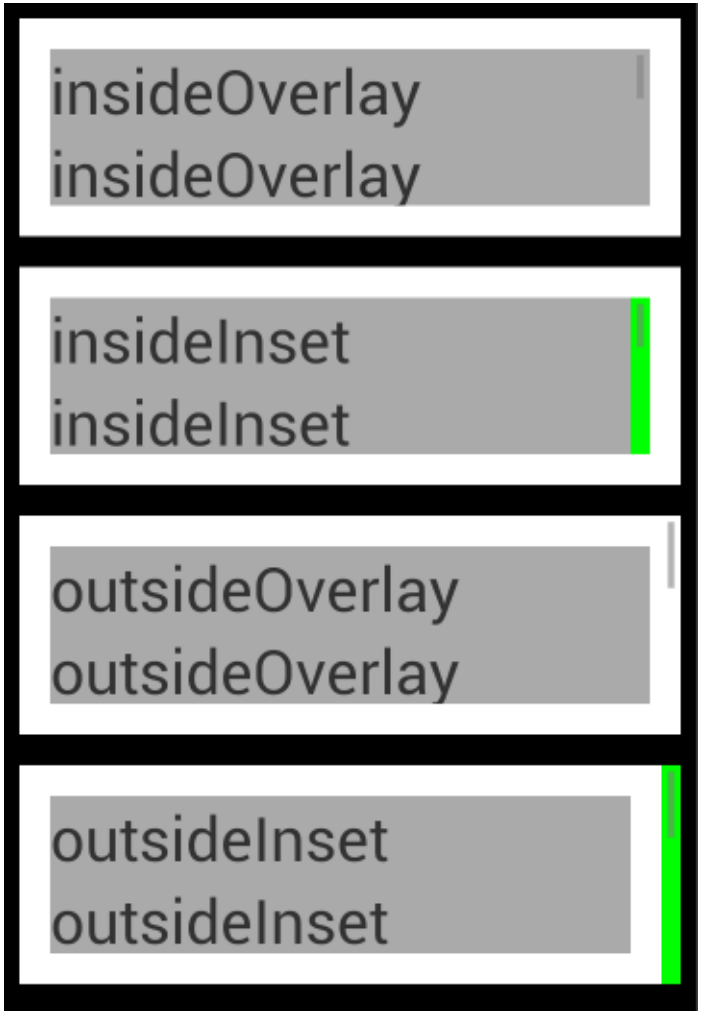
- FrameLayout
- GridLayout
- TableLayout (like simplified HTML table)

ScrollView

- Contains and scrolls one child
  - ScrollView should use match_parent or explicit size or layout_weight
  - Child should use wrap_content
  - Check lint warning

- Add gutter for scroll bar
  - fadeScrollbars="false"

scrollbarStyle



Inside margin; overlay content

Inside margin;
squeeze content to make room

Outside content; overlay margin

Outside content;
squeeze everything to make room